

System Architecture Diagrams

1. High-Level Flowchart

User → Frontend UI → API Gateway → Backend Services → Database

2. Reservation Module Flow

Create Reservation → Validate Input → Check Availability → Save Reservation → Return Confirmation

3. City Graph Module Interaction

Load Graph → Add Cities → Add Connections → Calculate Paths → Return Route

4. Class Diagram (Simplified)

- ReservationService
- createReservation()
- validate()
- save()

- CityGraph
- addCity()
- connectCities()
- shortestPath()

5. Module Interaction Diagram

Frontend → ReservationController → ReservationService → ReservationRepository → Database